

SDITO LAB

Name: SANJIB GHARA

Roll No.: CSE/23/065

Exam Roll No. 11600223092

Assignment 07

Problem Statement : Hosting a website on EC2

Aim

To launch an Amazon EC2 instance, configure a web server, and host a static website using terminal-based configuration.

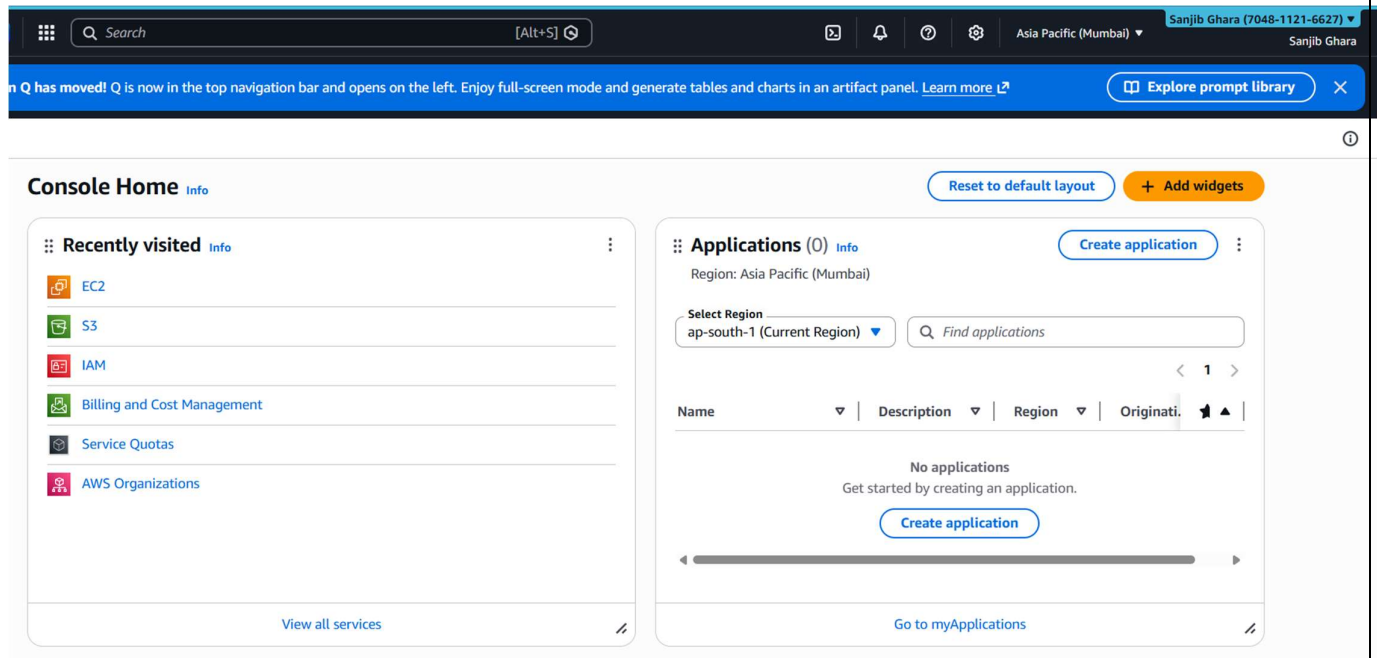
Requirements

- AWS Account
- Key pair (.pem file)
- HTML file (index.html)
- Internet connection
- Bitwise SSH Client

Solution:

Step 1: Launch an EC2 Instance

1. Login to AWS Management Console.



2. Search for EC2 and open the service.

This screenshot shows the AWS Management Console search results for 'EC2'. The search bar at the top contains 'EC2'. The left sidebar shows a list of services, with 'EC2' selected. The main content area displays 'Services' and 'Features' for EC2. The 'Services' section lists 'EC2' (Virtual Servers in the Cloud), 'EC2 Image Builder' (A managed service to automate build, customize and deploy OS images), and 'Recycle Bin' (Protect resources from accidental deletion). The 'Features' section lists 'EC2 Instances' (CloudWatch feature), 'EC2 Resource Health' (CloudWatch feature), and 'Dashboard' (EC2 feature). On the right, the 'EC2 cost' section shows the date range as 'Past 6 months', the region as 'Global', and the credits remaining as '\$139.99 USD'. The days remaining are '140 (July 6, 2026)'. The cost is shown as '0'.

This screenshot shows the AWS Management Console 'EC2' page. The left sidebar shows the 'EC2' service selected, with a red circle around the 'Instances' link. The main content area displays the 'Security Groups (2)' page. The table lists two security groups: 'default' (VPC ID: vpc-02f5c07b934db25fd) and 'launch-wizard-1' (VPC ID: vpc-02f5c07b934db25fd). Below the table, there is a 'Select a security group' section. At the bottom, the 'Instances' page is visible, showing a red circle around the 'Launch instances' button. The 'Instances' page also shows a table with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. The table is empty, and a message states 'No instances. You do not have any instances in this region.' Below the message is a 'Launch instances' button.

3. Click Launch instance. Instance Configuration

4. Enter:

- Name: Sanjib-first-hosting

5. Select:

- AMI: Ubuntu (Free Tier Eligible)

aws Search [Alt+S]

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices. [Take a walkthrough](#) [Do not show me this message again.](#)

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0317b0f0a0144b137

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

aws Search [Alt+S]

EC2 > Instances > Launch an instance

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type [Free tier eligible](#)

ami-019715e0d74f695be (64-bit (x86)) / ami-0848881f2a3dcebd1 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Summary

Number of instances: 1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

6. Click Create new key pair:

- Name: Sanjib-65
- Type: RSA
- Format: .pem

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-02f5c07b934db25fd

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-019715e0d74f695be

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Create key pair [X](#)

Key pair name

Key pairs allow you to connect to your instance securely.

sanjib-65

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Warning When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

7. Download and save the key file securely.
8. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet** and **Allow HTTP traffic from the internet** and click on **Launch instance** button.

The screenshot shows the AWS Management Console interface for launching an instance. The 'Network settings' tab is selected, and the 'Firewall (security groups)' section is expanded. A red circle highlights the 'Create security group' button and the three checked rules: 'Allow SSH traffic from', 'Allow HTTPS traffic from the internet', and 'Allow HTTP traffic from the internet'. A red arrow points to the 'Launch instance' button in the 'Summary' tab on the right.

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-02f5c07b934db25fd

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary [Info](#)

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Configure storage [Info](#) [Advanced](#)

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

[Click refresh to view backup information](#)

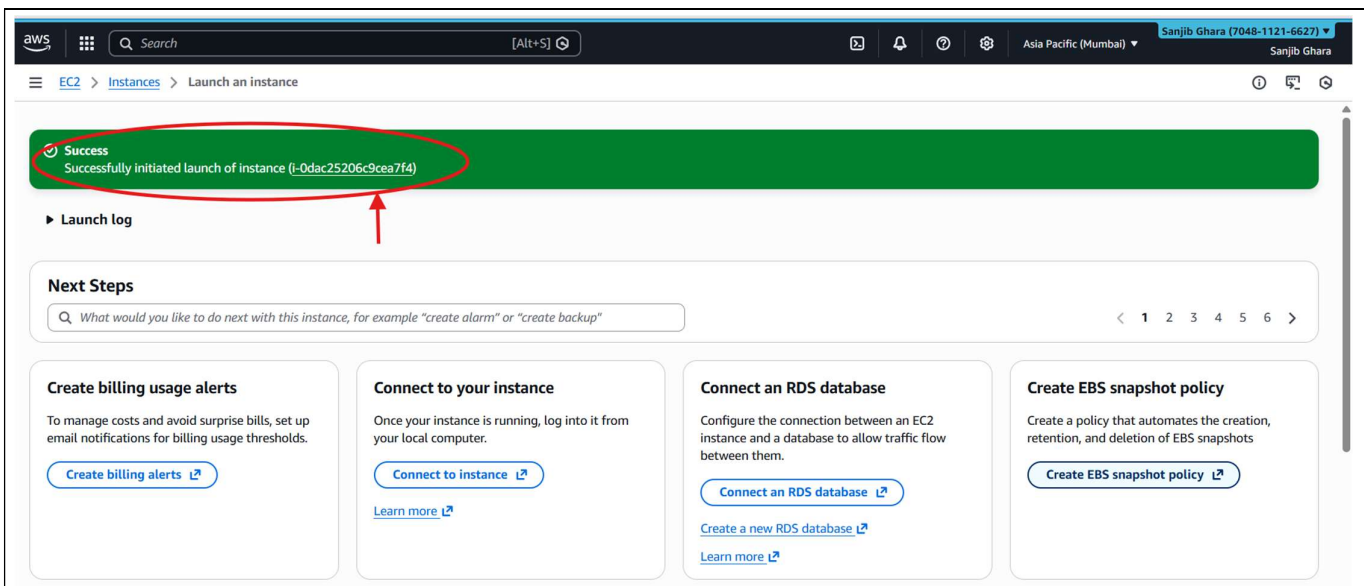
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

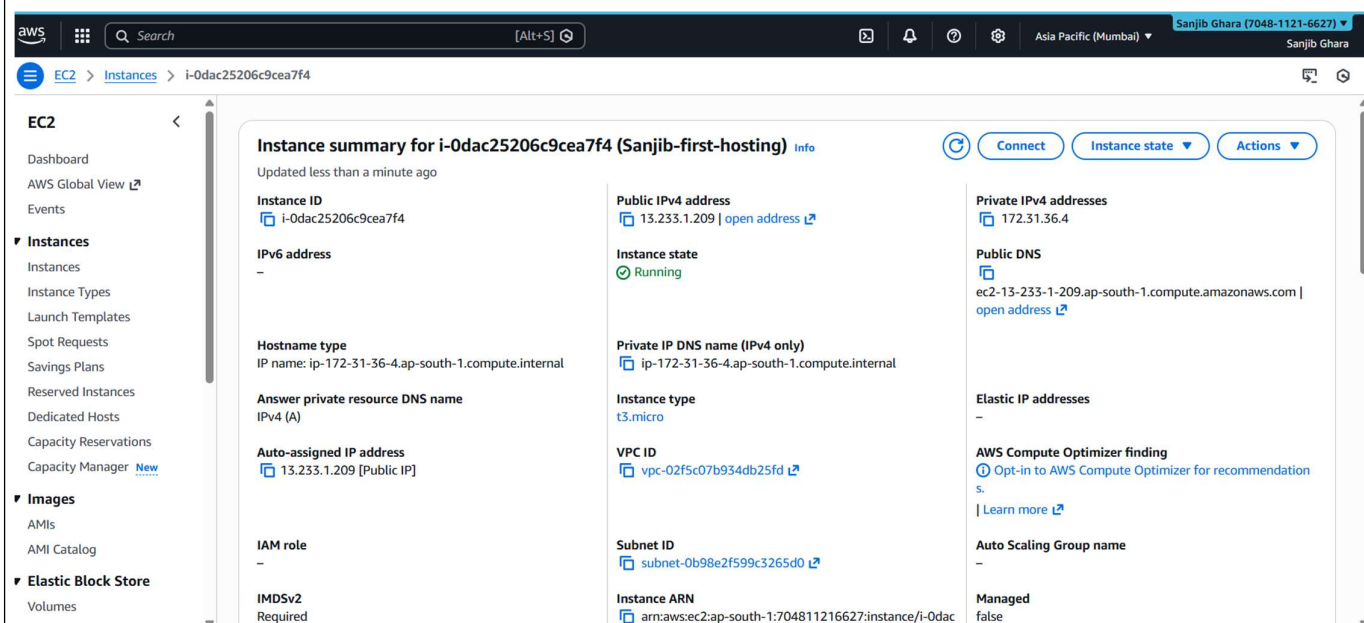
Advanced details [Info](#)

[Cancel](#) [Launch instance](#) [Preview code](#)

9. The Instance will be successfully created , return to the **Instance** page.



10: The Instance will be seen running, click on the newly created instance

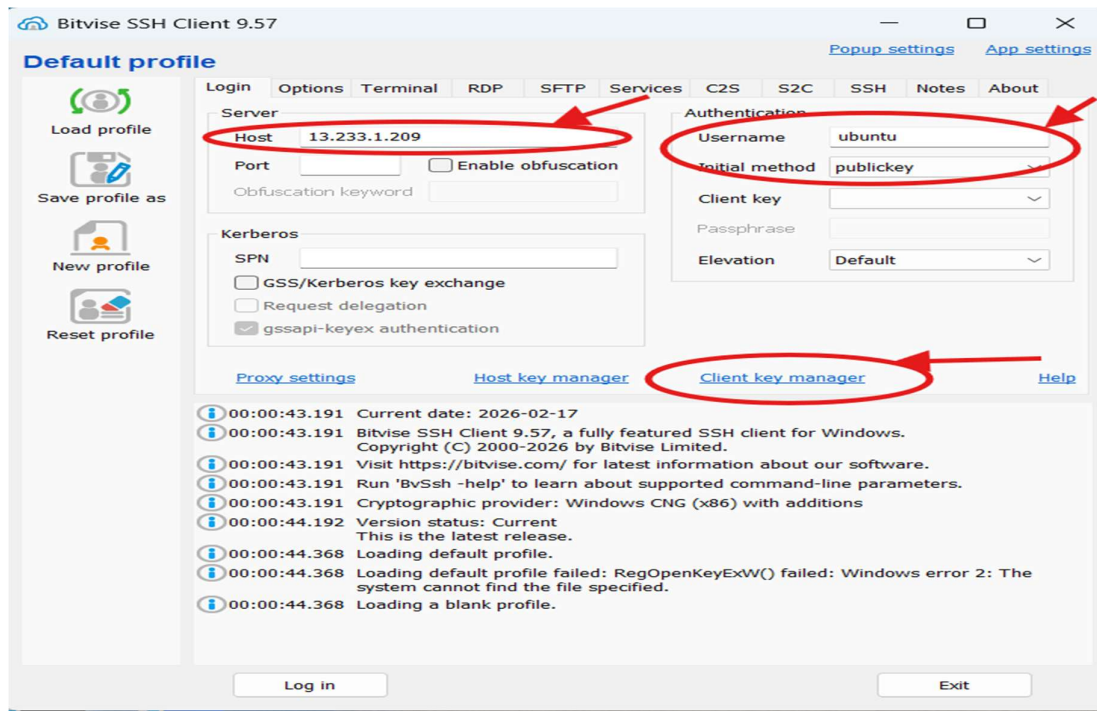


Step 2: Copy Public IPv4 Address

After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

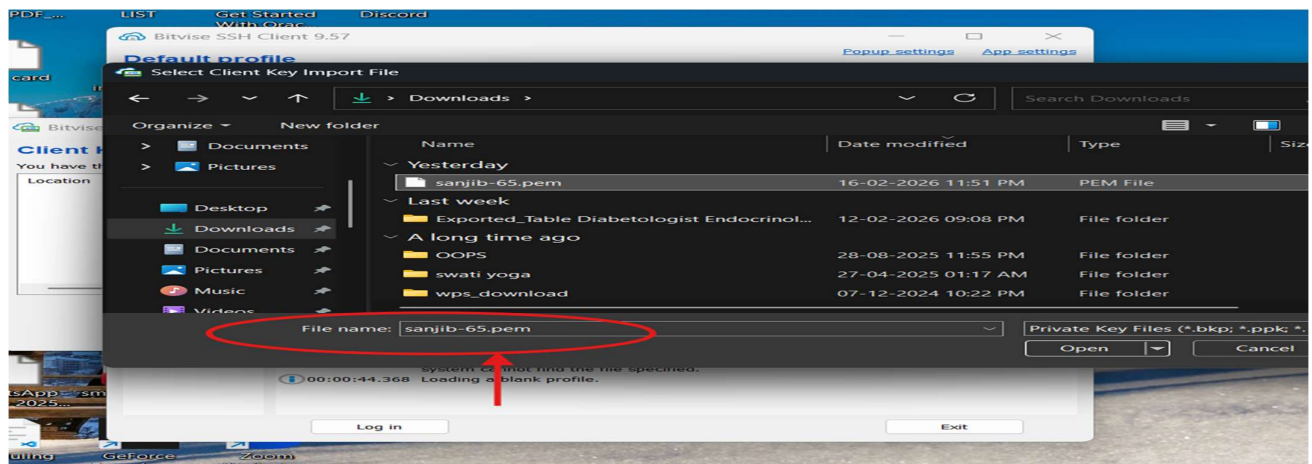
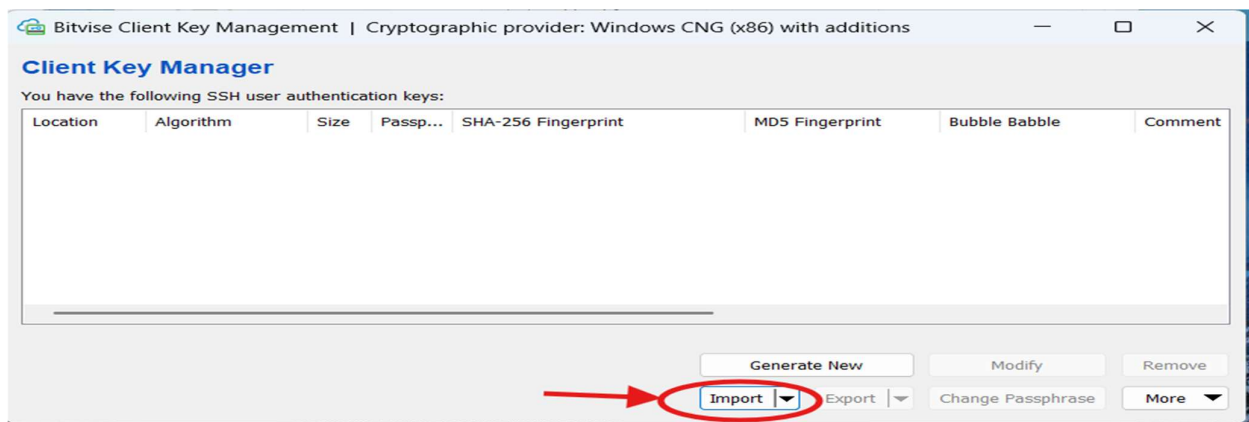
Step 3: Open Bitwise SSH Client and Configure Connection

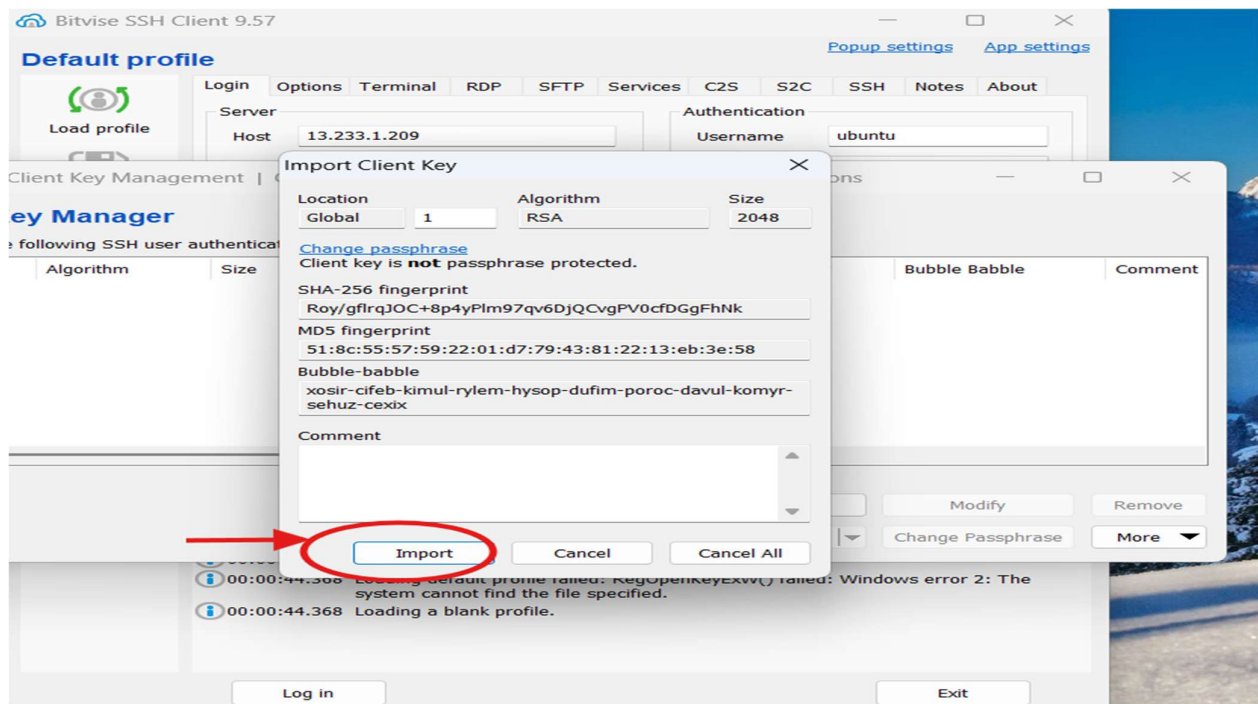
1. Open Bitwise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.
3. Enter ubuntu in the Username field.
4. Set Initial method to publickey.
5. Click on Client key manager



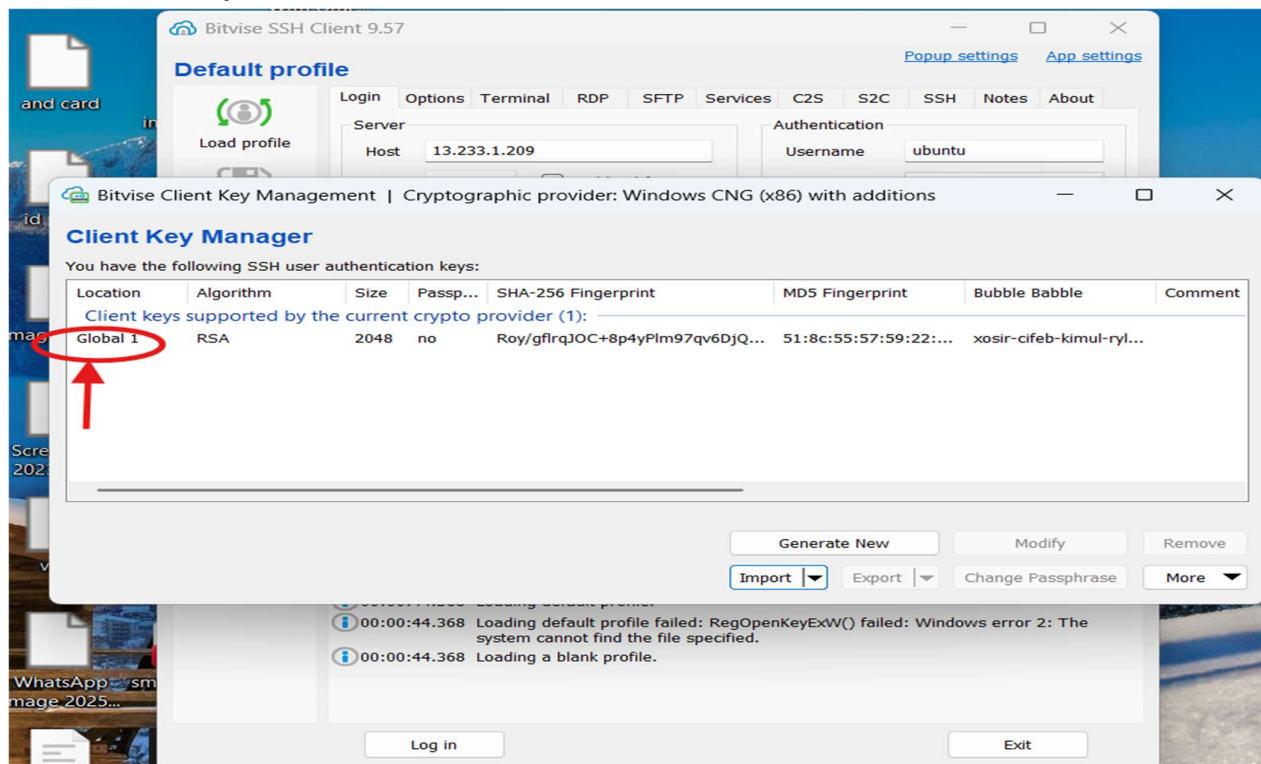
Step 4: Import Private Key

1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.



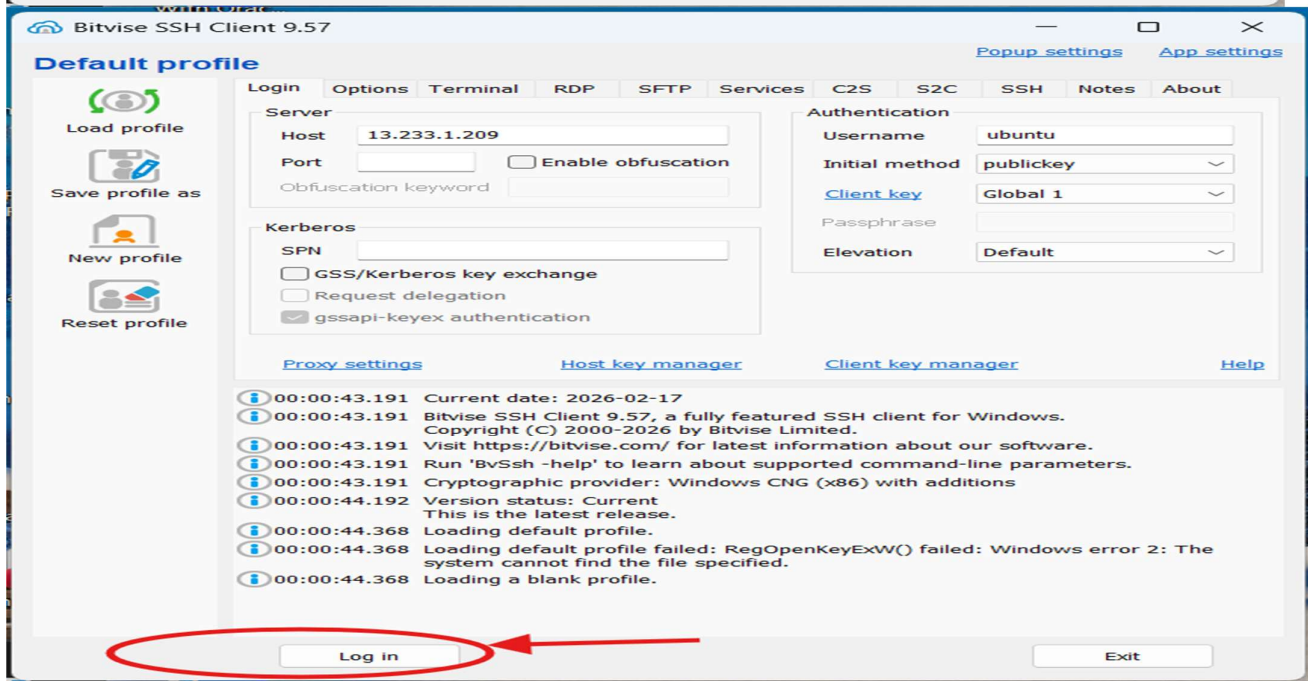
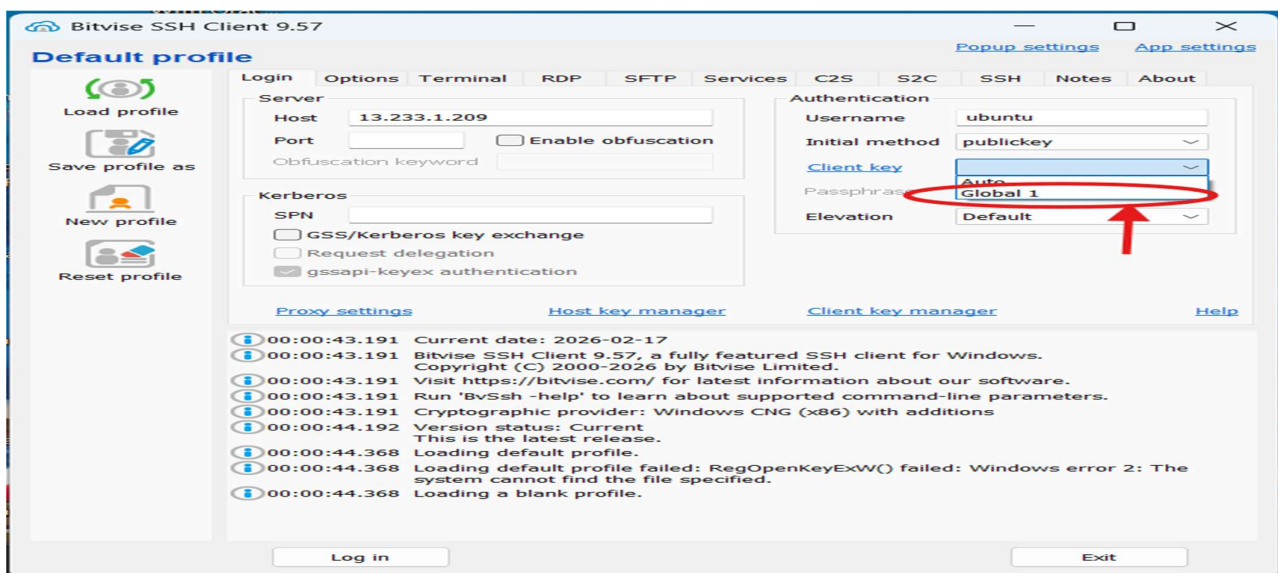


3. Save the key.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.



Step 6: Open Terminal Console

After successful login:

- Click on **New Terminal Console** from the **Bitvise** side panel.

```

ubuntu@13.233.1.209:22 - Bitvise xterm - ubuntu@ip-172-31-36-4: ~
Last login: Mon Feb 16 18:45:12 2026 from 152.58.182.78
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-36-4:~$ sudo apt-get update

```

Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

```

Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 39.9 MB in 6s (6529 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-36-4:~$ sudo apt-get upgrade

```

sudo apt-get update && sudo apt-get upgrade -y

Step 8: Install Nginx Web Server

Install Nginx using the command:

```
User sessions running outdated binaries:
```

```
ubuntu @ session #2: sshd[1061,1172]
```

```
ubuntu @ user manager service: systemd[1066]
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
ubuntu@ip-172-31-36-4:~$ sudo apt-get install nginx
```

```
sudo apt-get install nginx -y
```

```
Suggested packages:
```

```
fcgiwrap nginx-doc ssl-cert
```

```
The following NEW packages will be installed:
```

```
nginx nginx-common
```

```
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
```

```
Need to get 565 kB of archives.
```

```
After this operation, 1596 kB of additional disk space will be used.
```

```
Do you want to continue? [Y/n] y
```

Step 9: Set Permissions for Web Directory

Navigate to the web directory:

```
cd /var/www
```

```
ubuntu@ip-172-31-36-4:/home$ cd..
```

```
cd..: command not found
```

```
ubuntu@ip-172-31-36-4:/home$ cd ..
```

```
ubuntu@ip-172-31-36-4:/$ ls
```

```
bin          boot  etc    lib          lib64      media  opt   root  sbin          snap  sys  usr  
bin.usr-is-merged  dev  home  lib.usr-is-merged  lost+found  mnt    proc  run   sbin.usr-is-merged  srv   tmp  var
```

```
ubuntu@ip-172-31-36-4:/$ cd var
```

```
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ ls
```

```
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
```

```
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ cd www/
```

```
ubuntu@ip-172-31-36-4:/var/www$ ls
```

```
html
```

```
ubuntu@ip-172-31-36-4:/var/www$
```

Give proper permissions to the HTML directory:

```
sudo chmod 777 html/
```

```
ubuntu@ip-172-31-36-4:/var/www$ ls
```

```
html
```

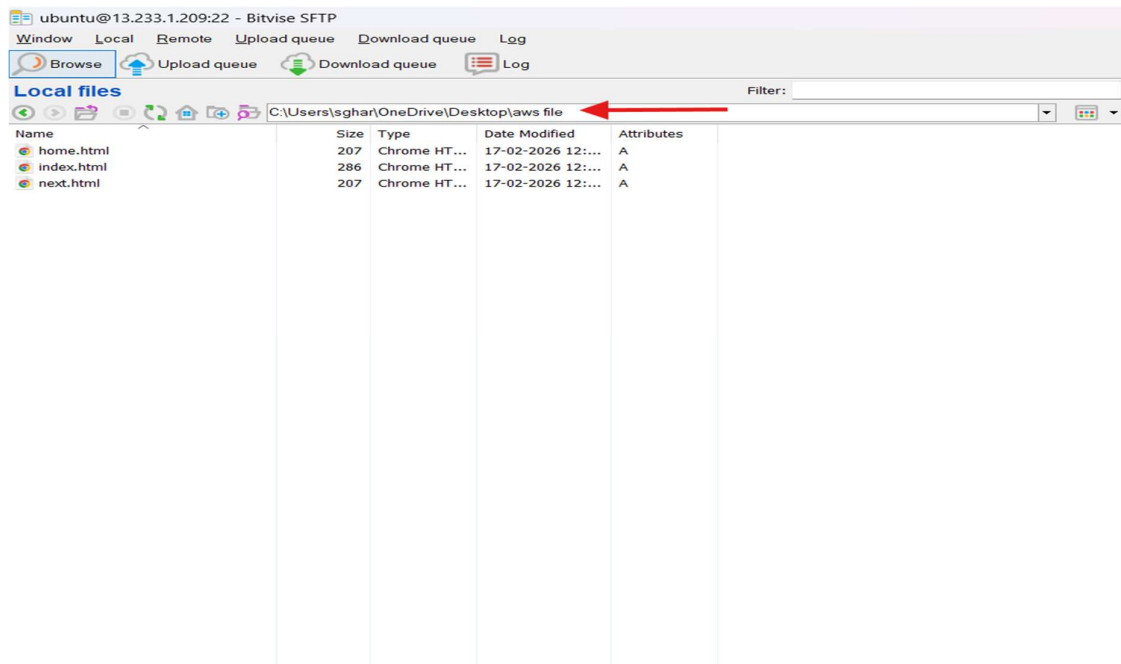
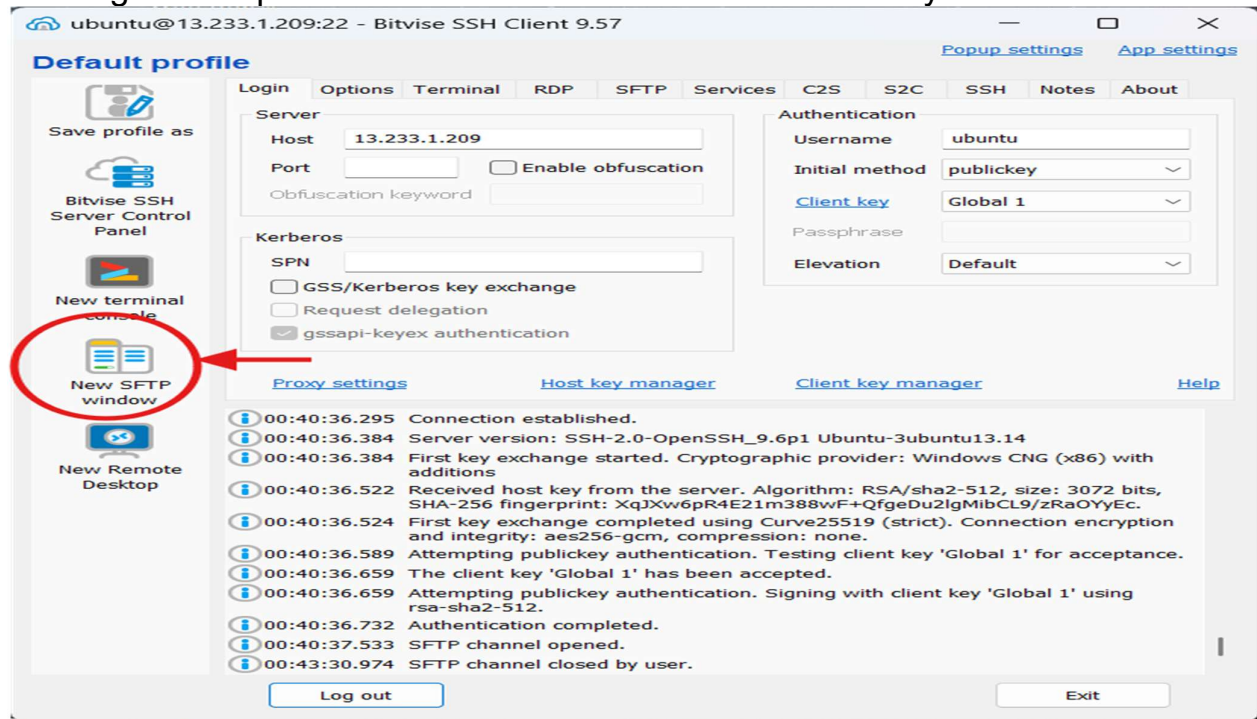
```
ubuntu@ip-172-31-36-4:/var/www$ sudo chmod 777 html
```

```
ubuntu@ip-172-31-36-4:/var/www$
```

Step 10: Upload Website Files Using SFTP

1. Click New SFTP window in Bitvise.
2. In the left pane (Local Files), navigate to the folder containing:
 - index.html
 - prev.html
 - next.html

3. In the right pane (Remote Files), navigate to: `/var/www/html`
4. Drag and drop all three files from local to remote directory.



Remote files Filter:

/home/ubuntu

Name	Size	Type	Date Modified	Permissions	Owner	Group
.cache	4,096	File folder	17-02-2026 12:...	drwx-----	ubuntu	ubuntu
.ssh	4,096	File folder	16-02-2026 11:...	drwx-----	ubuntu	ubuntu
.bash_history	139	BASH_HIST...	17-02-2026 12:...	-rw-----	ubuntu	ubuntu
.bash_logout	220	Bash Logou...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.bashrc	3,771	Bash RC So...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.profile	807	Profile Sou...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.sudo_as_admin_successful	0	SUDO_AS_...	17-02-2026 12:...	-rw-r--r--	ubuntu	ubuntu

Remote files Filter:

/home

Name	Size	Type	Date Modified	Permissions	Owner	Group
ubuntu	4,096	File folder	17-02-2026 12:...	drwxr-x---	ubuntu	ubuntu

Remote files Filter:

/

Name	Size	Type	Date Modified	Permissions	Owner	Group
bin	7	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
bin.usr-is-merged	4,096	File folder	26-02-2024 06:...	drwxr-xr-x	root	root
boot	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
dev	3,300	File folder	16-02-2026 11:...	drwxr-xr-x	root	root
etc	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
home	4,096	File folder	16-02-2026 11:...	drwxr-xr-x	root	root
lib	7	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
lib.usr-is-merged	4,096	File folder	08-04-2024 08:...	drwxr-xr-x	root	root
lib64	9	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
lost+found	16,384	File folder	12-12-2025 03:...	drwx-----	root	root
media	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
mnt	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
opt	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
proc	0	File folder	16-02-2026 11:...	dr-xr-xr-x	root	root
root	4,096	File folder	16-02-2026 11:...	drwx-----	root	root
run	1,020	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
sbin	8	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
sbin.usr-is-merged	4,096	File folder	31-03-2024 02:...	drwxr-xr-x	root	root
snap	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
srv	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
sys	0	File folder	16-02-2026 11:...	dr-xr-xr-x	root	root
tmp	4,096	File folder	17-02-2026 12:...	drwxrwxrwt	root	root
usr	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
var	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root

Remote files Filter:

Address: /var

Name	Size	Type	Date Modified	Permissions	Owner	Group
backups	4,096	File folder	22-04-2024 06:...	drwxr-xr-x	root	root
cache	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
crash	4,096	File folder	12-12-2025 03:...	drwxrwsrwt	root	root
lib	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
local	4,096	File folder	22-04-2024 06:...	drwxrwsr-x	root	staff
lock	9	File folder	12-12-2025 03:...	lrwxrwxrwx	root	root
log	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	syslog
mail	4,096	File folder	12-12-2025 03:...	drwxrwsr-x	root	mail
opt	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
run	4	File folder	12-12-2025 03:...	lrwxrwxrwx	root	root
snap	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
spool	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
tmp	4,096	File folder	17-02-2026 12:...	drwxrwxrwt	root	root
www	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
.updated	208	UPDATED File	12-12-2025 03:...	-rw-r--r--	root	root

Remote files Filter:

Address: /var/www

Name	Size	Type	Date Modified	Permissions	Owner	Group
html	4,096	File folder	17-02-2026 12:...	drwxrwxrwx	root	root

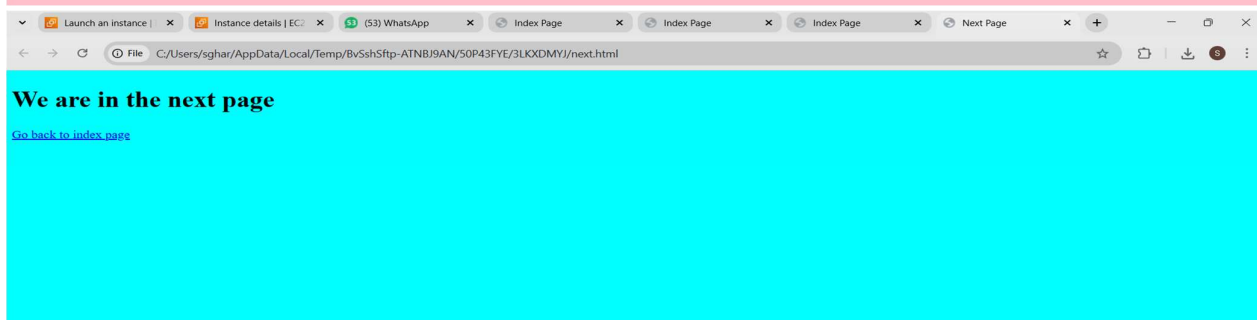
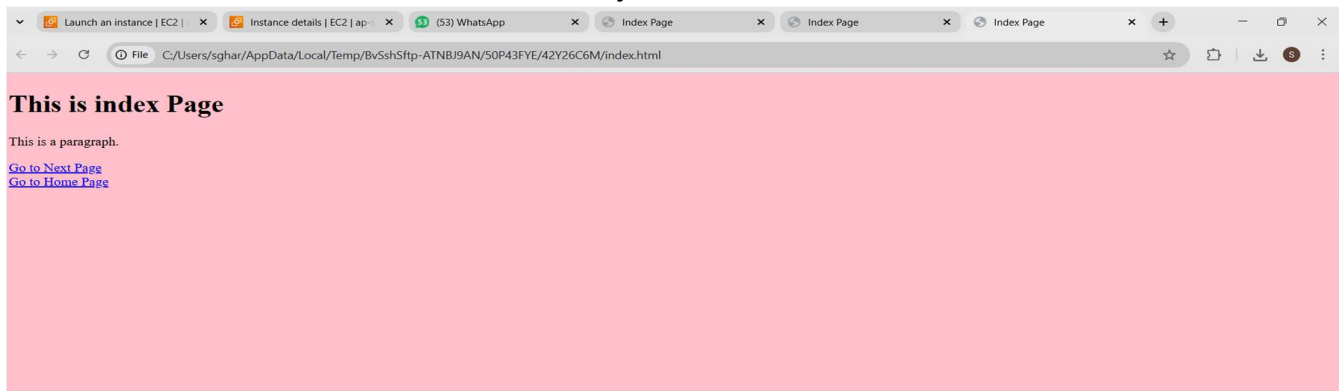
Local files					Remote files						
C:\Users\lgshan\OneDrive\Desktop\laws file					/var/www/html						
Name	Size	Type	Date Modified	Attributes	Name	Size	Type	Date Modified	Permissions	Owner	Group
home.html	207	Chrome HT...	17-02-2026 12:...	A	home.html	207	Chrome HT...	17-02-2026 12:...	-rwxr-xr-x	na	na
index.html	286	Chrome HT...	17-02-2026 12:...	A	index.html	286	Chrome HT...	17-02-2026 12:...	-rwxr-xr-x	na	na
next.html	207	Chrome HT...	17-02-2026 12:...	A	index.nginx-debian.html	615	Chrome HT...	17-02-2026 12:...	-rwxr-xr-x	root	root
					next.html	207	Chrome HT...	17-02-2026 12:...	-rwxr-xr-x	na	na

Step 11: Access the Website

Open a new browser tab and enter:

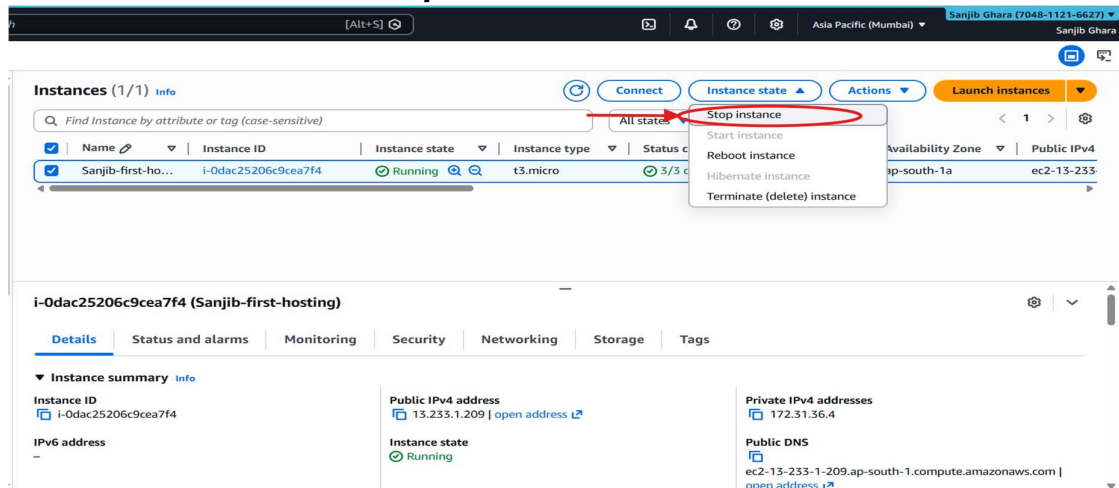
http://<Public-IPv4-address>

The static website will load successfully.

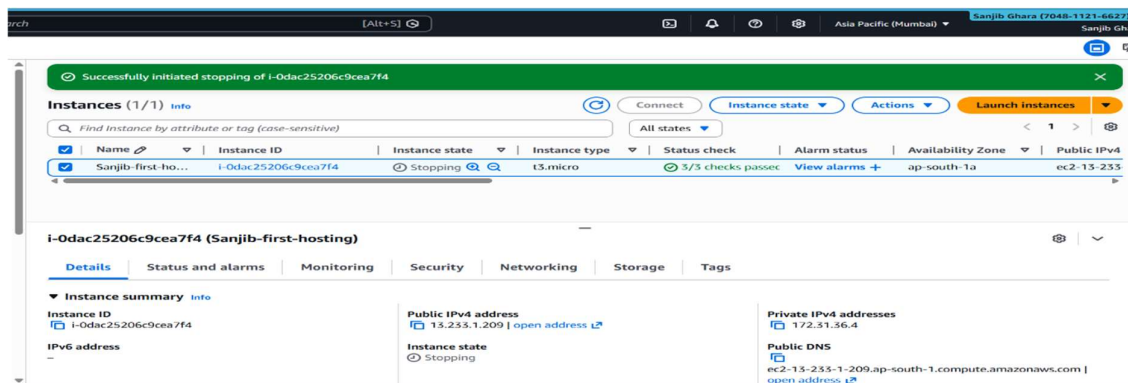
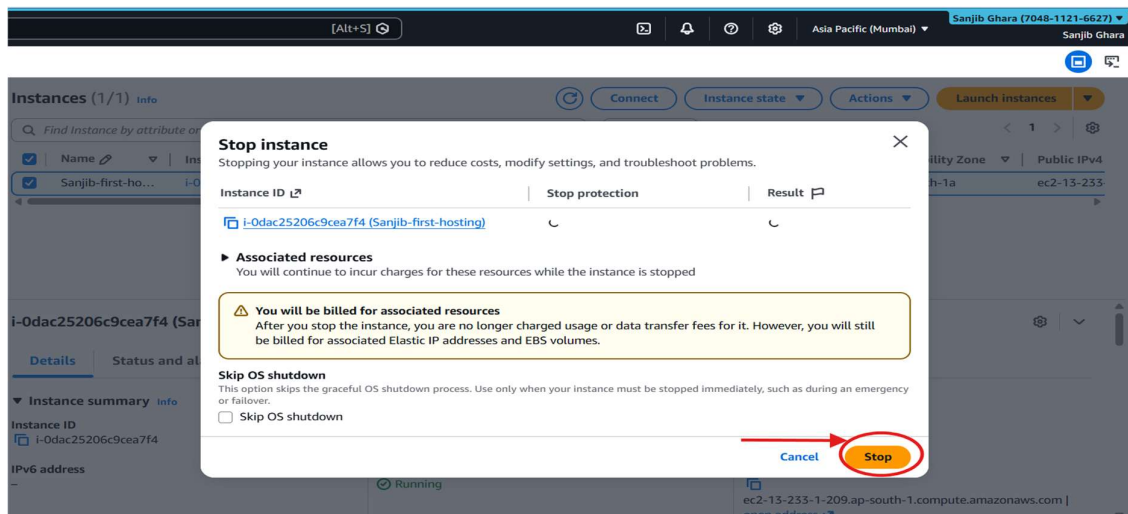


Step 12: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → **Stop instance**.

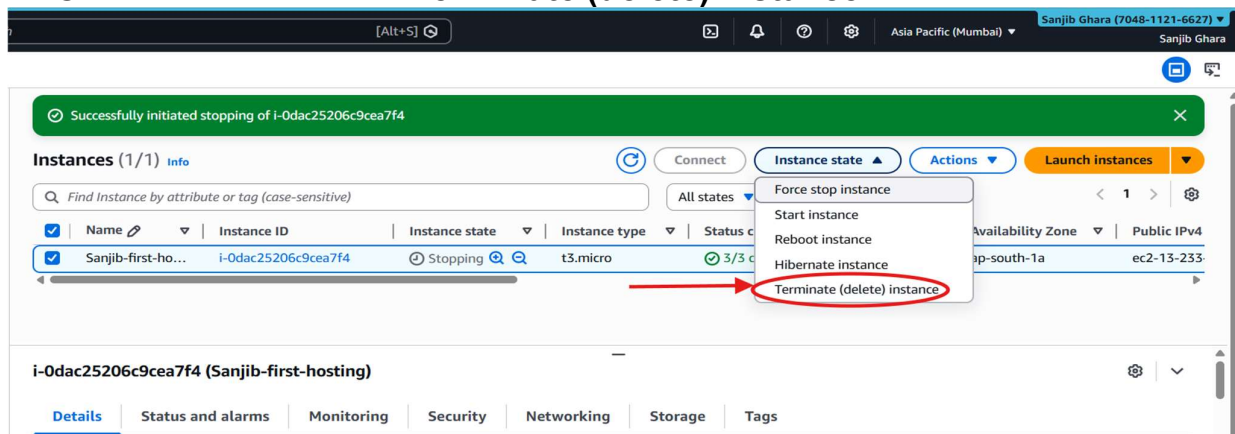


4. Confirm by clicking Stop.

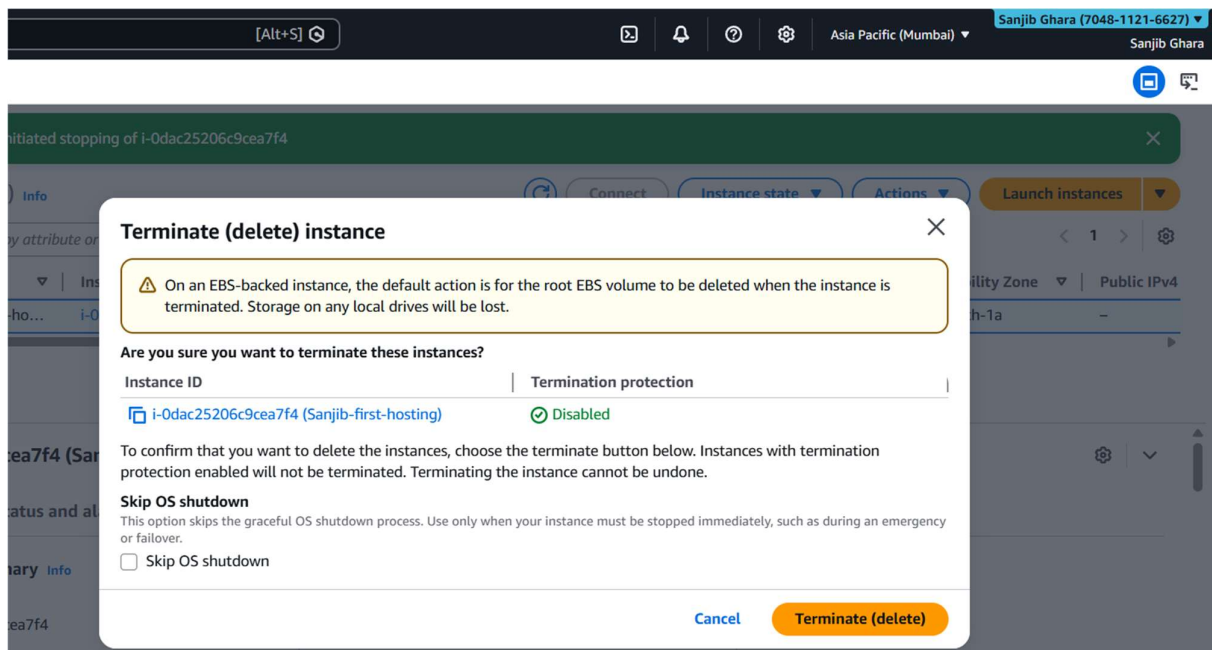


Step 13: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → **Terminate (delete) instance**.

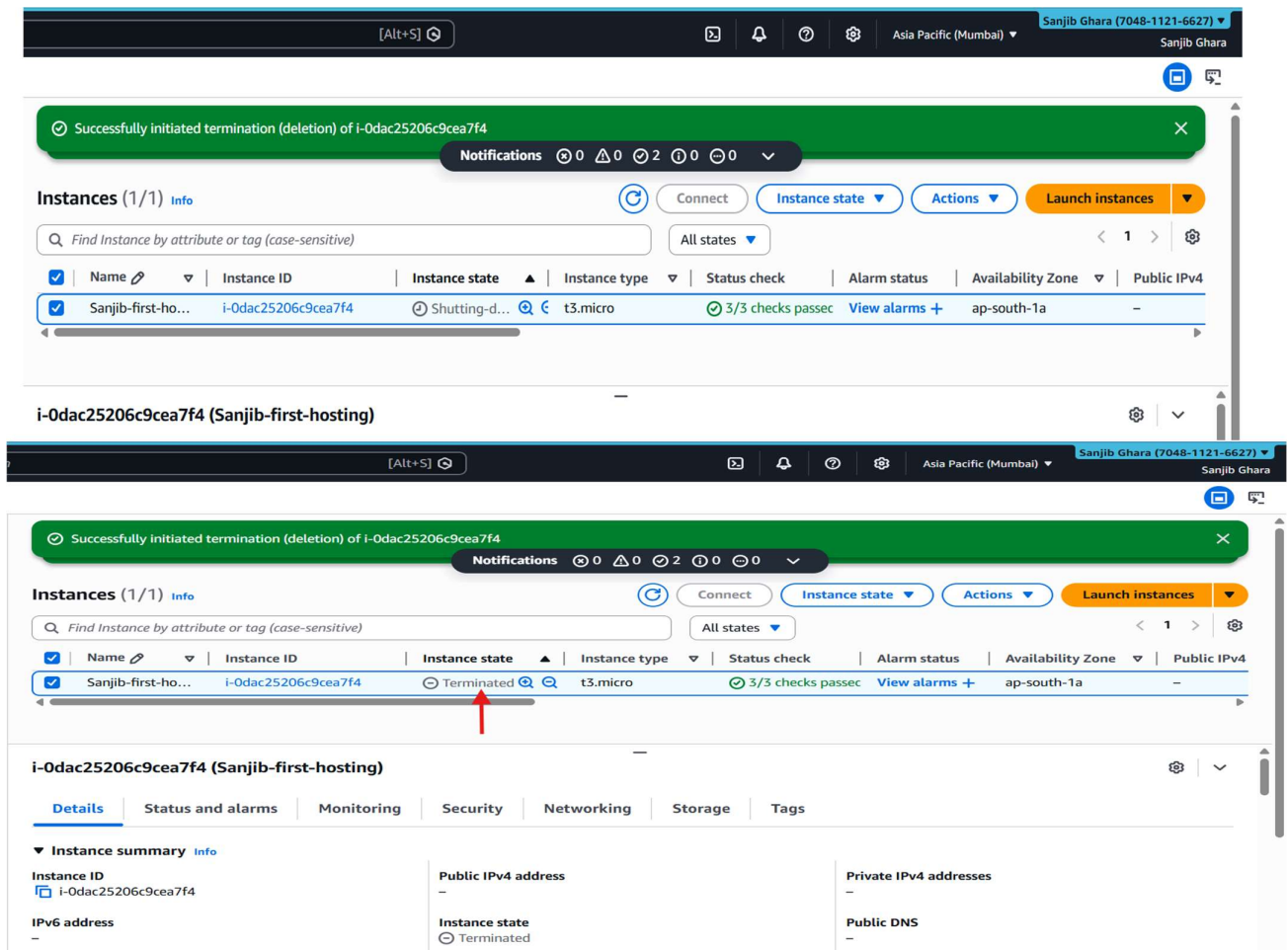


3. Confirm by clicking Terminate (delete).



Step 14: Instance Termination Confirmation

The EC2 instance is successfully terminated and all associated compute resources are released.



Conclusion

A static website consisting of multiple HTML files was successfully hosted on an Ubuntu-based EC2 instance using the Nginx web server and accessed via a public IPv4 address.